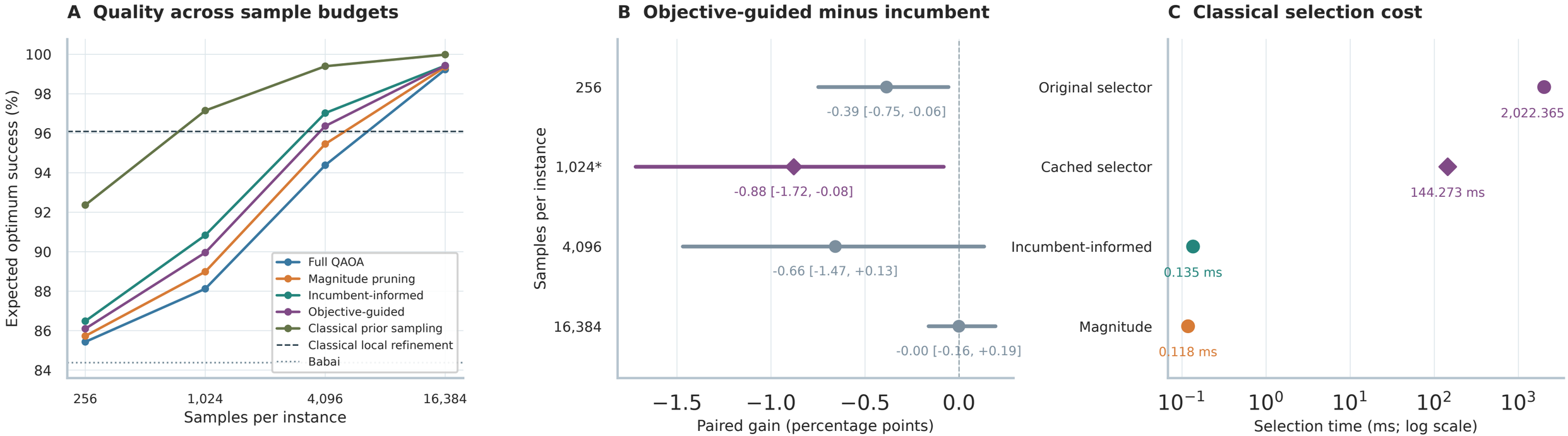
Fresh validation: solution quality and selection overhead

A–B: 256 fresh groups, ideal p = 1 QAOA. C: 32 separate fresh groups × 5 timing repetitions.



Cached implementation: 128/128 fresh masks and paths unchanged; 14.0× faster selection in this benchmark.

Paired 95% bootstrap intervals; *predeclared primary sample budget. Timing excludes shared preprocessing and compilation. No hardware execution.